

INHERITANCE

Aim:

To understand and implement inheritance concepts in Java.

PRE LAB EXERCISE

QUESTIONS

- ✓ What is inheritance?

Inheritance is a feature where one class **acquires the properties and methods of another class**.

- ✓ What is code reusability?

Code reusability means **using existing code again** instead of writing it multiple times.

- ✓ What is the use of extends keyword?

The **extends keyword** is used to **create a child class from a parent class** and reuse its code.

IN LAB EXERCISE

Objective:

To implement all types of inheritance.

PROGRAMS:

Student Result System (Single Inheritance)

Question:

A school wants to store student details and calculate marks. Create a base class Student and a derived class Result.

Code:

```
class Student {  
    String name;  
    int rollNo;
```

```
void getDetails() {  
    name = "Anitha";  
    rollNo = 101;  
}  
}  
  
class Result extends Student {  
    int marks = 85;  
  
    void display() {  
        System.out.println("Name: " + name);  
        System.out.println("Roll No: " + rollNo);  
        System.out.println("Marks: " + marks);  
    }  
}  
  
public class Main {  
    public static void main(String[] args) {  
        Result r = new Result();  
        r.getDetails();  
        r.display();  
    }  
}
```

Output:

Name: RAM

Roll No: 101

Marks: 85

```
Main.java 1 X
1  class Student {
2      String name;
3      int rollNo;
4
5      void getDetails() {
6          name = "Ranjith";
7          rollNo = 221;
8      }
9  }
10
11  class Result extends Student {
12      int marks = 85;
13
14      void display() {
15          System.out.println("Name: " + name);
16          System.out.println("Roll No: " + rollNo);
17          System.out.println("Marks: " + marks);
18      }
19  }
20
21  public class Main {
22
23  }
24  }
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE EXPLORER TERMINAL PORTS

```
PS C:\Users\ragur> & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:55933'
e-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\ragur\AppData\Local\Temp\vscodesws_6a209\jdt_ws\jdt.ls-java-project\bin' 'Main'
Name: Anitha
Roll No: 101
Marks: 85
PS C:\Users\ragur> ^C
PS C:\Users\ragur>
PS C:\Users\ragur> & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:63368'
e-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\ragur\AppData\Local\Temp\vscodesws_6a209\jdt_ws\jdt.ls-java-project\bin' 'Main'
Name: Ranjith
Roll No: 221
Marks: 85
PS C:\Users\ragur>
```

2. Bank Account System (Hierarchical Inheritance)

Question:

A bank has Savings and Current accounts. Both inherit from a common Account class.

Code:

```
class Account {

    void showAccountType() {

        System.out.println("Bank Account");

    }

}

class SavingsAccount extends Account {

    void interest() {

        System.out.println("Savings Account gives interest");

    }

}
```

```
class CurrentAccount extends Account {  
    void overdraft() {  
        System.out.println("Current Account supports overdraft");  
    }  
}
```

```
public class Main {  
    public static void main(String[] args) {  
        SavingsAccount s = new SavingsAccount();  
        CurrentAccount c = new CurrentAccount();  
  
        s.showAccountType();  
        s.interest();  
  
        c.showAccountType();  
        c.overdraft();  
    }  
}
```

Output:

Bank Account

Savings Account gives interest

Bank Account

Current Account supports overdraft

```
1 class Account { Main.java is a non-project file, only syntax errors are reported
5 }
6
7 class SavingsAccount extends Account {
8     void interest() {
9         System.out.println(x: "Savings Account gives interest");
10    }
11 }
12
13 class CurrentAccount extends Account {
14     void overdraft() {
15         System.out.println(x: "Current Account supports overdraft");
16    }
17 }
18
19 public class Main {
20     public static void main(String[] args) {
21         SavingsAccount s = new SavingsAccount();
22         CurrentAccount c = new CurrentAccount();
23
24         s.showAccountType();
25         s.interest();
26
27         c.showAccountType();
28         c.overdraft();
29     }
}
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE EXPLORER TERMINAL PORTS

```
PS C:\Users\ragur> & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:59273'
e-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\ragur\AppData\Local\Temp\vscodesws_9ab9f\jdt_ws\jdt.ls-java-project\bin' 'Main'
Bank Account
Savings Account gives interest
Bank Account
Current Account supports overdraft
PS C:\Users\ragur>
```

3. Vehicle System (Multilevel Inheritance)

Question:

A company classifies vehicles as Vehicle → Car → ElectricCar.

Code:

```
class Vehicle {
```

```
    void start() {
```

```
        System.out.println("Vehicle starts");
```

```
    }
```

```
}
```

```
class Car extends Vehicle {
```

```
    void fuelType() {
```

```
        System.out.println("Car uses petrol");
```

```
    }
```

```
}
```

```
class ElectricCar extends Car {  
    void battery() {  
        System.out.println("Electric car uses battery");  
    }  
}
```

```
public class Main {  
    public static void main(String[] args) {  
        ElectricCar e = new ElectricCar();  
        e.start();  
        e.fuelType();  
        e.battery();  
    }  
}
```

Output:

Vehicle starts

Car uses petrol

Electric car uses battery

```
1 class Vehicle {
2     void start() {
3         System.out.println(x: "Vehicle starts");
4     }
5 }
6
7 class Car extends Vehicle {
8     void fuelType() {
9         System.out.println(x: "Car uses petrol");
10    }
11 }
12
13 class ElectricCar extends Car {
14     void battery() {
15         System.out.println(x: "Electric car uses battery");
16    }
17 }
18
19 public class Main {
20     public static void main(String[] args) {
21         ElectricCar e = new ElectricCar();
22         e.start();
23         e.fuelType();
24         e.battery();
25     }
26 }
```

Bank Account
Current Account supports overdraft
PS C:\Users\ragur> ^C
PS C:\Users\ragur>
PS C:\Users\ragur> & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:61875' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\ragur\AppData\Local\Temp\vscodesws_9ab9f\jdt_ws\jdt.ls-java-project\bin' 'Main'
Vehicle starts
Car uses petrol
Electric car uses battery
PS C:\Users\ragur>

POST LAB EXERCISE

- ✓ Why Java does not support multiple inheritance using classes and how it is implemented?

One class inherits from **more than one parent class**.

- ✓ What is the role of the super keyword? Give examples.

The super keyword refers to the **parent class object**.

- ✓ Can a child class access private members of the parent class? Why?

No, a child class cannot directly access private members of the parent class.

Because:

- private members are accessible **only within the same class**.
 - They are not visible outside the class (even to child classes).
- ✓ Explain why hybrid inheritance is not supported in Java.

Java supports multiple inheritance using **Interfaces**.

Result:

Thus the different types of inheritance were implemented and executed successfully.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	
Faculty Signature		